

Resampling Techniques

Problem 7: Jackknife vs Infinitesimal Jackknife

We compare two variance estimation methods:

- Delete-1 Jackknife
- Infinitesimal Jackknife (IJ)

using logistic regression.

$$Y_i \sim \text{Bernoulli}(p_i), \quad \log \frac{p_i}{1 - p_i} = X_i^\top \beta$$

Logistic Regression Formulas

Log-likelihood:

$$\ell(\beta) = \sum_{i=1}^n \left[Y_i X_i^\top \beta - \log(1 + e^{X_i^\top \beta}) \right]$$

Score function:

$$U(\beta) = \sum_{i=1}^n X_i (Y_i - p_i)$$

Hessian:

$$H(\beta) = - \sum_{i=1}^n X_i X_i^\top p_i (1 - p_i)$$

Task 7.1: Fit Logistic Model

Example dataset:

```
from sklearn.datasets import load_wine
```

- 1 Convert outcome to binary (e.g., class 0 vs others).
- 2 Fit logistic regression.
- 3 Store $\hat{\beta}$ and $\hat{p}_i$.

Task 7.2: Delete-1 Jackknife

- ① For each $i = 1, \dots, n$:
 - Remove observation i
 - Refit logistic regression
 - Store $\hat{\beta}_{(-i)}$

- ② Compute:

$$\widehat{Var}_{jack} = \frac{n-1}{n} \sum (\hat{\beta}_{(-i)} - \bar{\beta})^2$$

Question: How computationally expensive is this?

Task 7.3: Infinitesimal Jackknife

Model:

$$\log \frac{p_i}{1 - p_i} = X_i^\top \beta, \quad p_i = \frac{e^{X_i^\top \beta}}{1 + e^{X_i^\top \beta}}$$

Score function:

$$U(\beta) = \sum_{i=1}^n X_i(Y_i - p_i)$$

Exact Hessian:

$$H(\beta) = \frac{\partial^2 \ell(\beta)}{\partial \beta \partial \beta^\top} = - \sum_{i=1}^n X_i X_i^\top p_i(1 - p_i)$$

Matrix form:

$$H(\beta) = -X^\top W X, \quad W = \text{diag}(p_i(1 - p_i))$$

Influence approximation:

$$\hat{\beta}_{(-i)} - \hat{\beta} \approx -H^{-1}X_i(Y_i - \hat{\rho}_i)$$

IJ variance:

$$\widehat{Var}_{IJ} = \sum (H^{-1}X_i(Y_i - \hat{\rho}_i)) (H^{-1}X_i(Y_i - \hat{\rho}_i))^{\top}$$

Comparison and Interpretation

Generate data using $p_i \in (0.1, 0.3, 0.5, 0.7, 0.9)$ and $n \in (30, 50, 100)$.

Create tables comparing :

- Delete-1 Jackknife SE
- Infinitesimal Jackknife SE
- True model SE.

for different values of p and n .

Discussion:

- When should IJ approximate jackknife well?
- Why might they differ for small n ?
- Which method scales better computationally?

Problem 8: Case, Residual and Wild Bootstrap

We study uncertainty of OLS estimator:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

using three bootstrap methods:

- Case Bootstrap
- Residual Bootstrap
- Wild Bootstrap

Dataset (Python Example):

```
from sklearn.datasets import fetch_california_housing  
data = fetch_california_housing(as_frame=True)
```

Task 8.1: Fit Regression Model

- ① Select one predictor (e.g., median income).
- ② Fit OLS model.
- ③ Store:
 - $\hat{\beta}_0, \hat{\beta}_1$
 - Residuals $\hat{\varepsilon}_i$
 - Fitted values $\hat{Y}_i$
- ④ Plot residuals vs X and comment on heteroskedasticity.

Task 8.2: Case Bootstrap

- 1 Resample pairs (X_i, Y_i) .
- 2 Refit regression for each bootstrap sample.
- 3 Store $\hat{\beta}_1^*$.
- 4 Estimate bootstrap standard error.
- 5 Construct 95% confidence interval. Use 2.5% and 97.5% bootstrap percentiles as lower and upper limit for Confidence intervals.

Discussion:

- What assumption does case bootstrap make about data?

Task 8.3: Residual Bootstrap

- ① Keep X_i fixed.
- ② Resample residuals $\hat{\varepsilon}_i$.
- ③ Generate:

$$Y_i^* = X_i^\top \hat{\beta} + \varepsilon_i^*$$

- ④ Refit regression.
- ⑤ Construct CI for β_1 . Use 2.5% and 97.5% bootstrap percentiles as lower and upper limit for Confidence intervals.

Question:

- When will this method fail?

Task 8.4: Introducing Heteroskedastic Data

Now we introduce a new regression model:

$$Y_i = 1 + 2X_i + \varepsilon_i$$

where

$$\varepsilon_i \sim N(0, 1 + X_i^2)$$

Thus,

$$\text{Var}(\varepsilon_i \mid X_i) = 1 + X_i^2$$

so the variance increases with X_i .

Dataset generation (Python):

```
import numpy as np
X = np.random.uniform(-2,2,n)
eps = np.random.normal(0, np.sqrt(1+X**2))
```

Task 8.5: Compare Bootstrap Methods Under Heteroskedasticity

Using the new simulated data:

- ① Fit OLS model.
- ② Plot residuals vs X (look for funnel pattern).
- ③ Apply:
 - Case Bootstrap
 - Residual Bootstrap
 - Wild Bootstrap
- ④ Construct 95% CI for β_1 using each method. Use 2.5% and 97.5% bootstrap percentiles as lower and upper limit for Confidence intervals.

Comparison and Interpretation

Create a table comparing:

- Standard errors
- CI lengths

Expected Observation:

- Residual bootstrap underestimates variability.
- Case bootstrap performs reasonably.
- Wild bootstrap adapts to changing variance.

Question: Why does residual bootstrap fail here?

Deliverable: Problem 8

- Tables comparing SE (Case / Residual / Wild) for both data sets.
- Histogram of bootstrap distributions
- Short explanation of differences